# MILITARY SPECIFICATION

## CHEESE, PIZZA BLEND

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers a specified blend of ready-to-use Italian-style cheeses for pizza pies for use by the Department of Defense as an item of general issue (see 6.1).

## 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

| | | |
|---|---|---|
| TT-C-495 | - | Coatings, Exterior, For Tinned Food Cans |
| PPP-B-636 | - | Boxes, Shipping, Fiberboard |
| PPP-C-29 | - | Canned Subsistence Items, Packaging and Packing of |
| PPP-C-96 | - | Cans, Metal, 28 Gage and Lighter |

MILITARY

| | | |
|---|---|---|
| MIL-L-1497 | - | Labeling of Metal Cans for Subsistence Items |

STANDARDS

MILITARY

| | | |
|---|---|---|
| MIL-STD-105 | - | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-129 | - | Marking for Shipment and Storage |
| MIL-STD-1162 | - | Sanitary Standards for Cheese and Cheese Products Plants |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.



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(Copies of specifications, standards and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

U.S. DEPARTMENT OF AGRICULTURE

U.S. Standards for Condition of Food Containers 7 CFR 2842

USDA General Specifications for Dairy Plants Approved for USDA Inspection and Grading Service

Dairy Plants, Surveyed and Approved for USDA Grading Service 7 CFR 2858

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug and Cosmetic Act, and Regulations Promulgated Thereunder

Cheese and Cheese Products - Definitions and Standards of Identity

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)



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3. REQUIRMENTS

3.1 Materials.

3.1.1 Cheese. Italian-style cheese shall include low moisture, part skim Mozzarella, Provolone, Parmesan and Romano, all of which shall conform to the Federal Standards of Identity for the applicable varieties of cheese, as promulgated by the Federal Food and Drug Administration (see 3.8). The low moisture, part skim Mozzarella cheese blend used in the blend formation shall not have been frozen prior to shredding.

3.2 Optional ingredients.

3.2.1 Mold inhibiting agents. Mold inhibiting agents, if used, shall consist of sorbic acid, potassium sorbate, sodium sorbate or any combination of two or more of these in an amount not to exceed 0.3 percent by weight of the finished product, calculated as sorbic acid.

3.2.2 Anti-caking agents. Anti-caking agents, if used, shall consist of silicon dioxide, calcium silicate, sodium silicoaluminate or any combination of two or more of these in an amount not to exceed 2.0 percent by weight of the finished product.

3.2.3 Spices. Spices, if used, shall consist of suitable flavoring substances other than any which, singly or in combination with other ingredients, simulate the flavor of cheese of any age or variety.

3.3 Preparation. Prior to shredding and grating, all varieties of the cheese (see 3.1.1) shall be properly cleaned so as to be free from all defects which are unwholesome or which will detract from the quality of the finished product.

3.4 Production.

3.4.1 Blend composition. The blend composition shall contain not less than 80 percent by weight of a combination of low moisture, part skim Mozzarella and Provolone cheese in shredded form and not more than 20 percent of a combination of Parmasan and Romano in grated form.

3.4.2 Shredding and grating. The Mozzarella and Provolone cheese shall be shredded into ribbons (shreds or shoe-strings style) with a nominal range in dimensions of approximately 1/2 inch to 2-1/2 inches (13 mm to 64 mm) in length and 1/8 inch (3 mm) to 3/8 inch (10 mm) in width, or in accordance with good commercial size. A preponderance of fines (except the identifiable gated portion of the blend) shall not be acceptable. In addition, the Parmesan and Romano components of the blend shall be grated uniformly into particle dimensions of approximately 1 by 3 millimeters.

3.5 Finished product.

3.5.1 Physical characteristics. The finished product shall comply with the following physical charateristics:



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(a) The color of the blended product may vary from a light straw shade to a medium yellow butter shade.

(b) The blended product shall be relatively free flowing from the container, uniformly blended, and show discernible shreds.

(c) There shall be no discernible free beads of moisture.

(d) The cheese texture shall be semi-soft and plastic and free from mealiness or crumbliness.

(e) The product shall have good melt down with flowing and stringy properties when exposed to normal baking temperatures used in pizza manufacture.

(f) The flavor and odor of the blended cheese when melted shall be pleasingly piquant and free from musty, metallic or oxidized, tallowy, soapy or unclean flavors and odors.

3.6 Plant qualification. The product shall originate, be processed, handled, and packaged in plants which are approved under the USDA General Specifications for Dairy Plants surveyed and approved for USDA Inspection and Grading Service, and listed by the US Department of Agriculture in the documents title "Dairy Plants Surveyed and Approved for USDA Grading Service" and approved for the finished product described in this specification or in plants meeting the requirements of MIL-STD-1162. Storage or warehouse facilities for the finished product for Military use shall meet the requirements of MIL-STD-1162.

3.7 Handling and delivery temperatures. The canned, vacuum packed product and flexible packages shall be promptly frozen after packaging and held at a temperature of 0°F (-18°C) during shipping and storage. The temperature at time of receipt shall not exceed 10°F (-12°C).

3.8 Delivery provisions. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the peformance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Plant qualification. The product shall be unacceptable if produced and stored in plants not meeting the requirements of 3.6.



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4.3 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.3.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.1.1 Examination of mold inhibiting agents, anti-caking agents, and spices. Determination of compliance of these optional ingredients with the identity requirements of 3.2.1 through 3.2.3, shall be made by examination of labels, invoices, or other valid documents. Non-compliance with the above referenced requirements shall be cause for rejection of the pertinent lot or the involved quantity of finished product made therefrom, as applicable.

4.3.1.2 Component testing. In addition to any testing required by 4.3.1, components shall be tested for the characteristics listed in table I. Results shall be applicable to the lot average. Nonconformance to the tin plate requirement shall be cause for rejection of the lot.

TABLE I. Component tests

| Component | Sample unit | Lot size expressed in | Inspection level | Characteristics | Results reported to nearest | Test ref and reqmt para |
|---|---|---|---|---|---|---|
| Cans and lids 1/ | 1 can w/lid | Cans | S-1 | Tin plate | 0.01 lb/base box (0.1 $g/m^2$) | 4.4.1 and 5.1.1 |

1/ Exterior can coating, when required, and can label shall be tested in accordance with the applicable subsidiary specifications except that tests shall be performed on the same cans submitted for tin plate testing with additional cans as required to complete the sampling plan of TT-C-495.

4.3.2 In-process examination.

4.3.2.1 Preparation, formulation, shredding, and grating. Examination shall be made during processing to determine compliance with the requirements of 3.2.1 and 3.2.2 as concerns specified component weight limitations when applicable, of 3.3 and 3.4 as concerns procedures and formulation, and of 3.7 as concerns freezing after packaging. Formulation records shall be maintained. Noncompliance with one or more processing requirements, reflected by actual examination or by records, shall be cause for rejection of the involved quantity of product.



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4.3.3 End item inspection.

4.3.3.1 Net weight examination. The filled and sealed primary containers shall be examined for the defects listed in table II. The lot size shall be expressed in units of primary containers. The sample unit shall be one primary container. The inspection level shall be S-3 and the AQL shall be 2.5 expressed as percent defective.

TABLE II. Net weight tests 1/

| Category | Defect |
|---|---|
| Minor | |
| 201 | Less than 63 ounces (1786 g) (for cans) 2/ |
| 202 | Less than 3.8 ounces (108.0 g) (for flexible unit packs) 3/ |

1/ Lot shall be rejected if sample data indicate that lot average net weight is less than specified net weight.

2/ Report to the nearest 1/4 ounce (1 g).

3/ Report to the nearest 0.1 ounce (0.1 g).

4.3.3.2 Product examination. The cheese blend shall be examined for the defects listed in table III. The lot size shall be expressed in units of primary containers. The sample unit shall be the contents from one sealed primary container. The inspection level shall be S-2 and the AQL shall be 1.5 major A and B and 6.5 minor defects per hundred units. The finding of any major A defect shall be cause for rejection of the lot.

TABLE III. Cheese blend tests

| Category | | | Defect |
|---|---|---|---|
| Major A | Major B | Minor | |
| 101 | | | Presence of foreign material, e.g., metal particles, insect or insect parts, wood, paint, glass, etc. |
| 102 | | | Presence of foreign or objectionable flavor or odor such as musty, metallic, oxidized, tallowy, soapy, unclean, etc. (After melting; see 3.5.1) |
| 103 | | | Presence of moldy areas. |



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TABLE III. Cheese blend tests (cont'd)

| Category | | | Defect |
|---|---|---|---|
| Major A | Major B | Minor | |
| | 151 | | Size of shreds, ribbons, shoe strings, and grated components not as specified. 1/ |
| | 152 | | Presence of free beads of moisture. |
| | 153 | | Mealy or crumbly pieces or texture not plastic (not applicable to grated cheese components). |
| | 154 | | Meltdown, flowing, and stringy properties not as specified. 2/ |
| | 155 | | Flavor and odor not pleasingly piquant after meltdown. |
| | | 201 | Pieces are so compacted together so as to impede relatively free flow from container. |
| | | 202 | Color not as specified. |
| | | 203 | Not uniformly blended. |

1/ Measure six pieces taken from each sample unit.

2/ Examine approximately 1/2 to 1 ounce (14 to 28 g) sample from each sample unit. Place on aluminum foil or flat pyrex baking dish (special oven ware) and proceed with melting as specified in 3.5.1 (e).

4.3.4 Packaging inspection.

4.3.4.1 Flexible unit pack examination. The flexible unit packs, when required, shall be examined for the defects listed in table IV. The lot size shall be expressed in units of flexible unit packs. The sample unit shall be one filled and sealed flexible unit pack. The inspection level shall be I and the AQL shall be 0.25 major A, 1.5 major B, and 6.5 minor defects per hundred units.

TABLE IV. Flexible unit pack tests

| Category | | | Defect |
|---|---|---|---|
| Major A | Major B | Minor | |
| 101 | | | Tear or hole through unit pack. |
| | 151 | | Not clean |



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TABLE IV. Flexible unit pack tests (cont'd)

| Category | | | Defect |
|---|---|---|---|
| Major A | Major B | Minor | |
| 102 | | | Closure not sealed |
| | 153 | | Nomenclature missing, incorrect, or illegible. |
| | | 201 | Other required label information missing, incorrect, or illegible. |

4.3.4.2 Examination of cans for vacuum. Examinations shall be made in accordance with the applicable examination criteria contained in PPP-C-29.

4.3.4.3 Examination of the filled and sealed shipping container. Examination of the filled and sealed shipping container shall be in accordance with the examination criteria contained in the appendix to PPP-B-636. In addition, the following defects shall be included in the table of examination: Major: National stock number, item description, contract number, or date of pack markings missing, incorrect or illegible; Minor: (1) Other required markings missing, incorrect or illegible. (2) Arrangement not as specified. When applicable for level B pack, containers reinforced with other than nonmetallic strapping or pressure-sensitive adhesive, filament reinforced tape shall not be acceptable. For level C pack, examination shall be in accordance with the examination criteria of the aforementioned specification except that only the defects applicable to marking, closure (see 5.2.2), and maximum marked net weight of 48 pounds (21.8 kg) shall apply.

4.3.4.4 Examination for leakage of cans. Examination for leakage of cans shall be made in accordance with the applicable examination criteria of PPP-C-29.

4.3.4.5 External examination of cans. External examination of cans shall be performed in accordance with the applicable requirements of the United States Standards for Condition of Food Containers.

4.3.5 Examination for temperature at time of delivery. The product shall be examined for temperature at time of delivery to determine compliance with the requirements of 3.7. The sample unit shall be one shipping container. Lot size shall be expressed in terms of the sample unit, and the sample size for inspection shall be the number of sample units indicated by inspection level S-2. Alternatively, if the shipment is accompanied by suitable temperature recording devices of proven accuracy and spaced in appropriate locations to perit representative readings, such temperature records shall be used to determine compliance with the aforementioned referenced requirements. Finding of any temperature reading not in accordance with the requirements shall be cause for rejection of the lot.



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4.4 Methods of inspection.

4.4.1 Tin coating weights. Tin coating weights shall be determined by any method specified in PPP-C-96.

5. PACKAGING

5.1 Preservation. The product shall be preserved in accordance with level B or C as specified (see 6.1).

5.1.1 Level B. A net weight of 64 ounces (1814 g) of product shall be unit packed into an open-top style, round metal can, size 603 by 700, with a soldered or welded side seam and compound-lined, double-seamed ends. A net weight tolerance of minus 1 ounce (28 g) is permitted, provided the average net weight of cans in any lot is not less than 64 ounces (1814 g). The can shall be made throughout from not less than 0.25 pound per base box (5.6 $g/m^2$) electrolytic tinplate (see 4.4.1). The can shall have a sufficient base plate weight and temper to protect the product during shipment and storage. The can shall be coated overall on the outside with a coating conforming to type I of TT-C-495. The filled can shall be sealed under a vacuum of not less than 25 inches of mercury (16.6 kPa) (see 4.3.4.2) and shall not leak when examined in accordance with 4.3.4.4.

5.1.2 Level C.

5.1.2.1 Cans. The product shall be unit packed as specified in 5.1.1 except that cans with or without commercial exterior coating will be acceptable.

5.1.2.2 Flexible unit packs. A net weight of 4 ounces (113.4 grams) of product shall be placed into a flexible unit pack which shall protect the product from deterioration or damage from the supply source to the first consuming activity. A net weight tolerance of minus 0.2 ounces (5.4 g) is permitted, provided the average net weight of flexible unit packs in any lot is not less than 4 ounces (113 g).

5.2 Packing. The product shall be packed in accordance with level B or C as specified (see 6.1 and 6.2).

5.2.1 Level B. Six cans of product, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed, closed and reinforced in accordance with style RSC, grade V3c, V3s or V4s of PPP-B-636. The cans shall be packed on end, 3 in length, 2 in width and 1 in depth within the container.

5.2.1.1 Alternate shipping container. When specified (see 6.1 and 6.4), the product shall be packed as specified in 5.2.1, except the box shall be fabricated of grade W5c or W5s fiberboard.

5.2.1.2 Nonmetallic reinforcement of shipping containers. When specified (see 6.1 and 6.3), the shipping container shall be reinforced with non-metallic strapping or pressure-sensitive adhesive, filament reinforced tape in accordance with the appendix of PPP-B-636. Two reinforcing bands shall be used, one lengthwise and one girthwise.



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5.2.2 Level C. Six cans of product or not more than 48 pounds (21.8 kg) of product in flexible unit packs, preserved as specified in 5.1, shall be packed in a manner to ensure carrier acceptance and safe delivery at destination at the lowest transporation rate for such supplies. The shipping container shall be in accordance with Uniform Freight Classification or National Motor Freight Classification, as applicable, except that the closure of fiberboard boxes shall be in accordance with method II as specified in the appendix of PPP-B-636.

5.3 Labeling.

5.3.1 Cans. Cans shall be labeled in accordance with MIL-L-1497. In addition, the following thawing instructions shall be included:

DIRECTIONS FOR THAWING

Place unit pack of frozen product in refrigerator (35°F to 45°F) (1.7 to 7°C) for 24 hours prior to using

5.3.2 Flexible unit packs. Flexible unit packs which have commercial labeling that complies with the Federal Food, Drug and Cosmetic Act and regulations promulgated thereunder will be acceptable. Each flexible unit pack shall show the same thawing instructions as specified for cans in 5.3.1.

5.4 Marking. In addition to any special marking required by the contract; shipping containers shall be marked in accordance with MIL-STD-129.

5.4.1 Precautionary marking. The following precautionary marking shall be printed or legibly stenciled in capital letters of the same size, not less than 1 inch (25 mm) high on top of each shipping container:

PERISHABLE - KEEP FROZEN (0°F) (-18°C) OR BELOW

6. NOTES

6.1 Ordering data.

(a) Title, number and date of this specification.
(b) Selection of applicable levels of preservation and packing (see 5.1, 5.2, 5.2.1 and 6.2).
(c) When non-metallic reinforcement is required for level B packing (see 5.2.1.2).
(d) When alternate shipping container specified in 5.2.1.1 is adequate (see 6.4).

6.2 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7. Level B packing specified in 5.2.1 is intended for transfer at sea operations or specific Navy overseas operations.



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6.3 Selection of nonmetallic reinforcement. The non-metallic reinforcement specified in 5.2.1.2 should be selected for shipping containers that are destined for transfer at sea or other specific overseas operations.

6.4 Use of alternative shipping container. Packing specified in 5.2.1.1 may be used when experience or knowledge of the destination and mode of shipment would indicate that a lower grade of fiberboard can be used.

6.5 Shelf life. It is essential to keep the canned product frozen at a temperature not to exceed 0F (-18C) for a shelf lift up to 12 months.

6.6 Award of contract for the product specified in this specification will be limited to plants known to maintain the required sanitation conditions of 3.6 (also see 4.2).

Custodians:

Army – GL
Navy – SA
Air Force – 50

Preparing Activity:

Army – GL

Project No. 8910-0419

Review activities:

Army – MD
Navy – MC, MS
DP – SS

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